
BIL24P01 - MULTI-SCALE MODELLING

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I am also deeply grateful to the **Bioinnovation Labs, PES University** for providing the opportunity to undertake this internship. The experience has been invaluable in broadening my knowledge of biotechnology and the intricate workings of the human body. The exposure to advanced modelling techniques and the role of technology in healthcare has highlighted the significant impact that scientific research and innovation can have on improving people's lives.

This internship has been a profound learning journey, and I am thankful for the chance to contribute to cutting-edge research in the field of biomedicine.

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ABSTRACT

This report details the collaborative efforts undertaken during our internship at the Bioinnovation Lab, focusing on the computational modelling of how lung cancer cells interact with T-cells according a designed pathway. Our work began with a foundational understanding of cellular biology, particularly the roles of proteins, signalling pathways, and T Cells in cancer immunology. We explored advanced modelling techniques, such as Boolean networks and agent-based modelling, to construct a multiscale model capable of simulating complex biological interactions.

The project specifically modelled the behaviour of lung cancer cell pathway using tools like WebMaBoSS for simulating gene and protein interactions and PhysiCell for modelling tumour microenvironment dynamics. Our findings highlight the importance of computational models in predicting cancer progression and evaluating potential therapeutic strategies, offering valuable insights for pharmaceutical development where direct testing is often impractical.

INTRODUCTION

Cells are the fundamental units of life, and their behaviour is regulated by complex biological processes. Proteins play diverse roles within cells, acting as enzymes, structural components, and signalling molecules. Signalling pathways are intricate networks of molecular interactions that control essential cellular functions such as growth, differentiation, and apoptosis.

T Cells are a vital component of the adaptive immune system, designed to detect and eliminate cells that are infected or cancerous. They achieve this through antigen recognition, where T-cell receptors (TCRs) bind to antigens presented by Major Histocompatibility Complex (MHC) molecules on antigen-presenting cells (APCs). This recognition triggers T Cell activation, leading to a targeted immune response against the cancer cells.

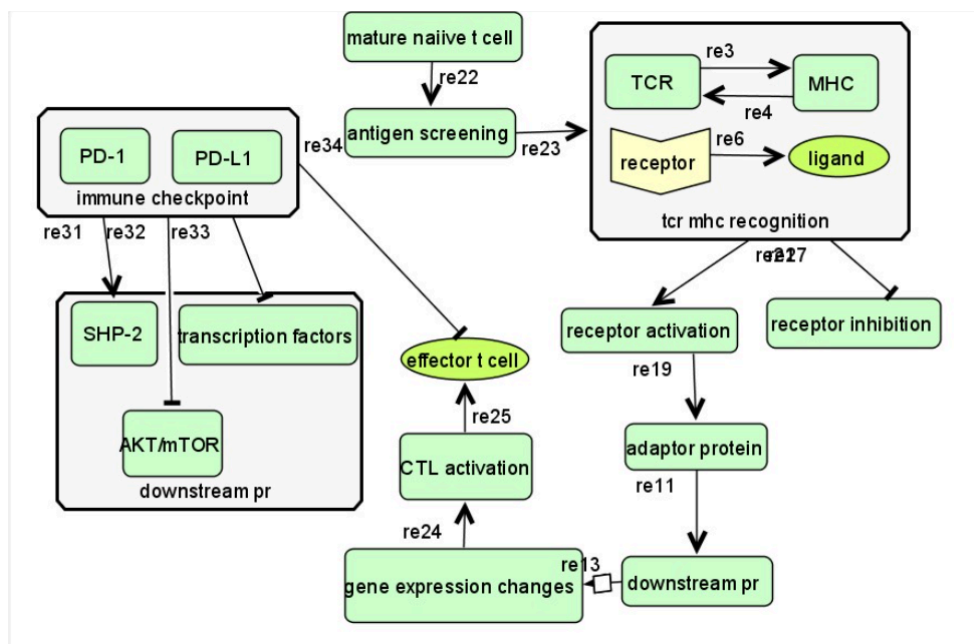
Cancer cells can exploit various mechanisms to evade immune detection. These mechanisms may include altering surface antigens, down-regulating MHC molecules, or secreting immunosuppressive factors. Specifically, our focus is on lung cancer cells with mutations that allow them to avoid detection by T Cells, making them particularly challenging to target with conventional therapies. We aim to first gain a thorough understanding of how T-cells interact with cancer cells and how other immune system elements assist in the same.

Given the limitations of direct testing and experimental approaches, computational models are increasingly vital in cancer research. They allow researchers to simulate complex biological systems and predict the effects of mutations and treatments. Our project leverages advanced modelling techniques to understand how different cells migrate and the rate kinetics driving cancer cell death.

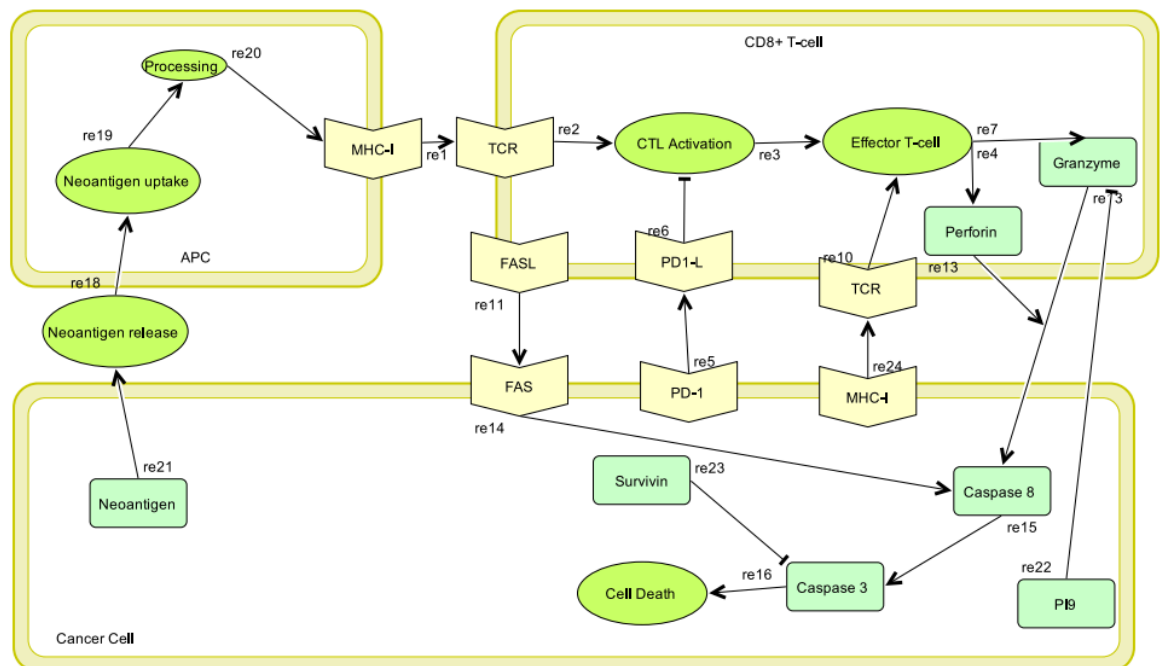
By using tools such as WebMaBoSS & PhysiCell, we seek to gain insights into tumour behaviour, interactions within the tumour microenvironment, and responses to potential treatments. The ultimate goal is to enhance our understanding of cancer progression and contribute to the development of more effective therapeutic strategies.

MATERIALS AND METHODS

1. Literature Review: To gain a general understanding of how the pathways operate, a comprehensive review of various research papers was done, mainly focusing on how T cells operate in tumour cells and their role in activating or inhibiting various apoptotic pathways. Relevant data was extracted.
2. Pathway Construction: All the pathways were constructed using CellDesigner software, and mapped using positive and negative interaction legends to define the binary interactions between all the molecules involved, where each molecular species was represented as a node.



- The above pathway was constructed using the computer application CellDesigner and information from various research papers as well as referential YouTube videos that helped in compiling information. Multiple papers on T cell maturation, TCR-MHC complex interaction, PD-1,PD-L1 interaction as well as papers on how the cytotoxic T cell activation leads to an effector cell being formed were referred to. The interactions were represented using arrows with an arrowhead and a perpendicular head which represent activation and inhibition respectively. The individual pathways were connected after more literature survey to produce a combined pathway.



4. Deriving SBML Qual file: The .xml pathway file from CellDesigner is converted into a .sbml file (qual) using a Python module known as CasQ
5. WebMaBoSS simulation: The .sbml (qual) file is loaded to get a Boolean definition of the relationship between elements. A stochastic simulation of the Boolean network was conducted to understand the probabilistic states different nodes would take and was compared with evidence from the literature. A simulation of around 10000 cycles for 100 seconds was run, where the initial conditions were set as follows:

Rates	Initial values	Outputs	Parameters	Settings
				50%
CTLAActivation_simple_molecule				13%
Caspase3				13%
Caspase8				10%
CellDeath_simple_molecule				14.0000000
EffectorTcell_simple_molecule				19%
FAS				100%
FASL				8%
Granzyme				12%
MHCLAPC				100%
MHCLCancerCell				100%
Neoantigen				100%
Neoantigenrelease_simple_molecule				100%
Neoantigenuptake_simple_molecule				100%
PD1				49%
PD1L				52%
PI3				100%
Perforin				15%
Processing_simple_molecule				100%
Survivin				100%
TCR				11%

Rates Initial values Outputs Parameters Settings

time_tick	1	
max_time	100	
sample_count	10000	
discrete_time	0	
use_physrandgen	0	
seed_pseudorandom	0	
display_traj	0.5	
statdist_traj_count	0	
statdist_cluster_threshold	5	
thread_count	1	
statdist_similarity_cache_max_size	20000	

Rates Initial values Outputs Parameters Settings

CTLAActivation_simple_molecule	☒
Caspase3	☐
Caspase8	☐
CellDeath_simple_molecule	☒
EffectorTcell_simple_molecule	☒
FAS	☐
FASL	☐
Granzyme	☒
MHCIAPC	☐
MHCI_CancerCell	☐
Neoantigen	☒
Neoantigenrelease_simple_molecule	☒
Neoantigenuptake_simple_molecule	☒
PD1	☒
PD1L	☒
PIG	☐
Perforin	☒
Processing_simple_molecule	☐
Survivin	☐
TCR	☐

The obtained .bnd and .cfg files were downloaded which would be used for the PhysiCell simulation

5. PhysiCell simulation: The download .bnd and .cfg files were loaded onto Physicell in the Intracellular option of the Cell types panel. Three cell types were defined: Cancer cells, CD8⁺ T-cells and antigen-presenting cells. Following this was the setting of initial parameters (cell death rate, rules of interaction, adhesion, mechanical and diffusion properties, microenvironment, etc). The executable file for running the simulation was compiled prior to loading all the data. The microenvironment was basically hypoxic as cancer cells can proliferate in such an environment as they undergo Glycolysis to produce lactate to survive in such an environment. The initial conditions are as follows:

PhysiCell Studio: C:\Users\hipra\Desktop\PhysiCell\sample_projects\Multiscale\bin\PhysiCell_settings.xml

Studio File View Action Help

Config Basics Microenvironment Cell Types User Params Rules ICs Run Plot

Domain (micron)

Xmin	-500	Xmax	500	dx	20
Ymin	-500	Ymax	500	dy	20
Zmin	20	Zmax	20	dz	20

Times

Max Time	1440.0	min	1 d	0 h	0 min	0.0 min
Diffusion dt	0.2	min				
Mechanics dt	0.05	min				
Phenotype dt	2	min				

Misc runtime parameters

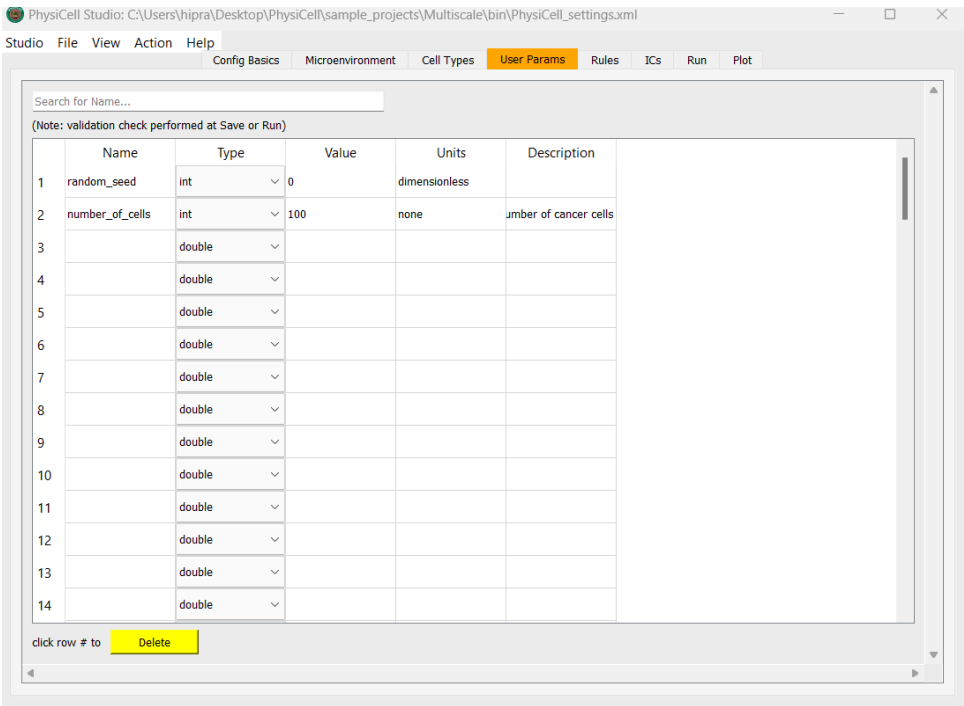
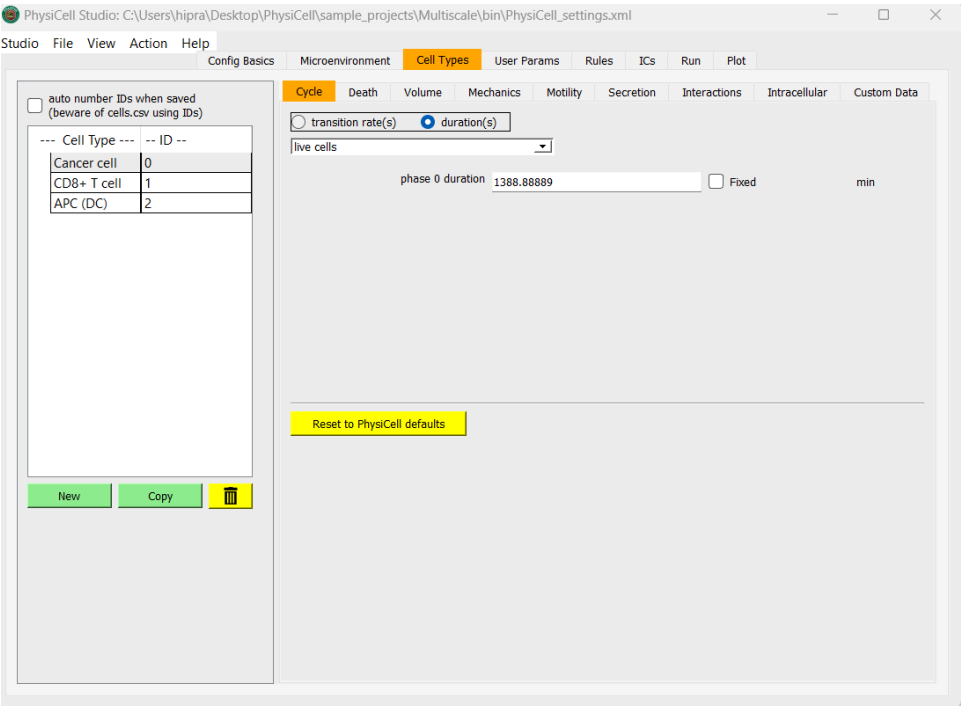
# threads	4		
output folder	output		
Save data (intervals):	☒ SVG 2 min	☒ Full 2 min	☒ Sync
Plot SVG substrate:	☐ enable lactate	☐ Limits enabled min max	

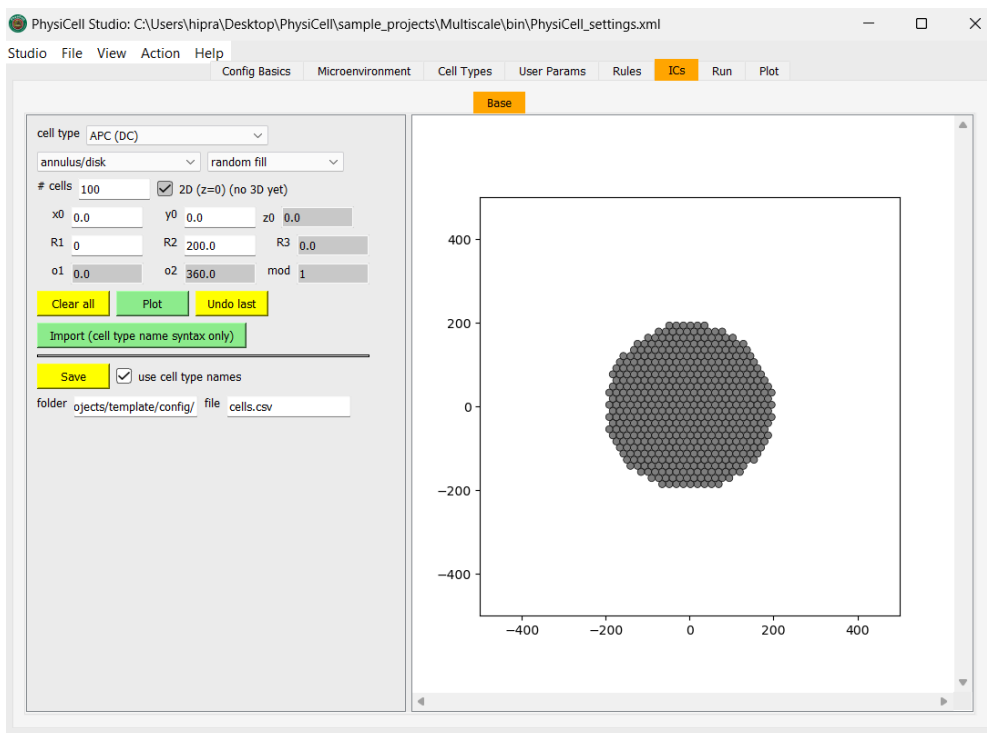
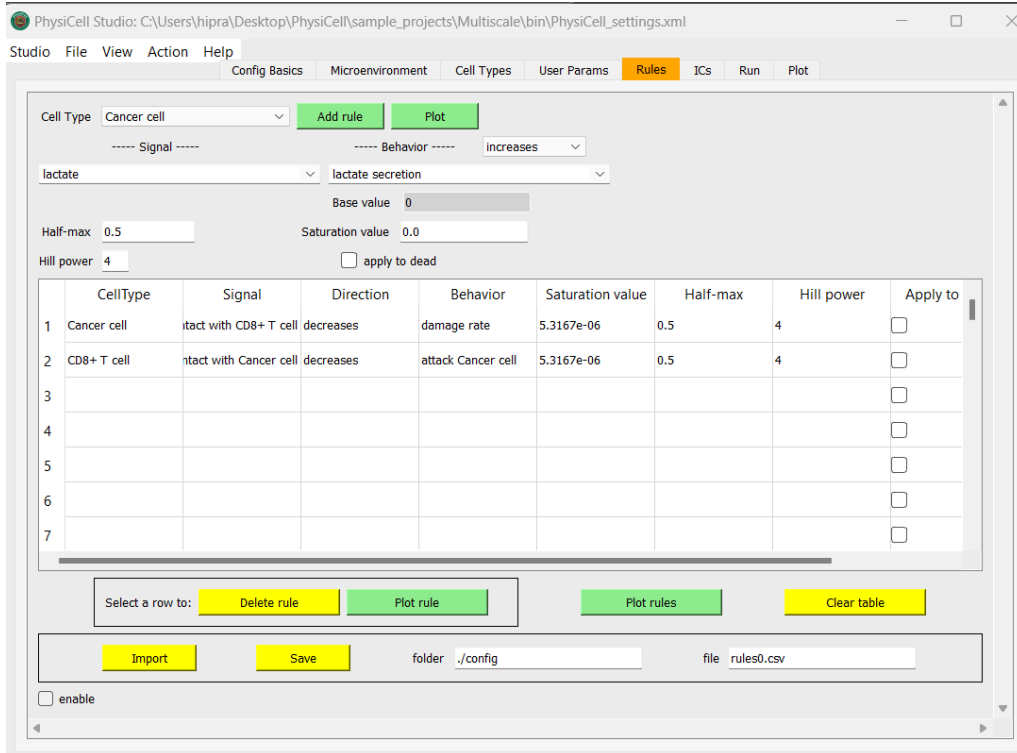
Initial conditions of cells (x,y,z, type)

☒ enable	folder	'sample_projects/template/config/	file	cells.csv	Import
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Cells' global behaviors

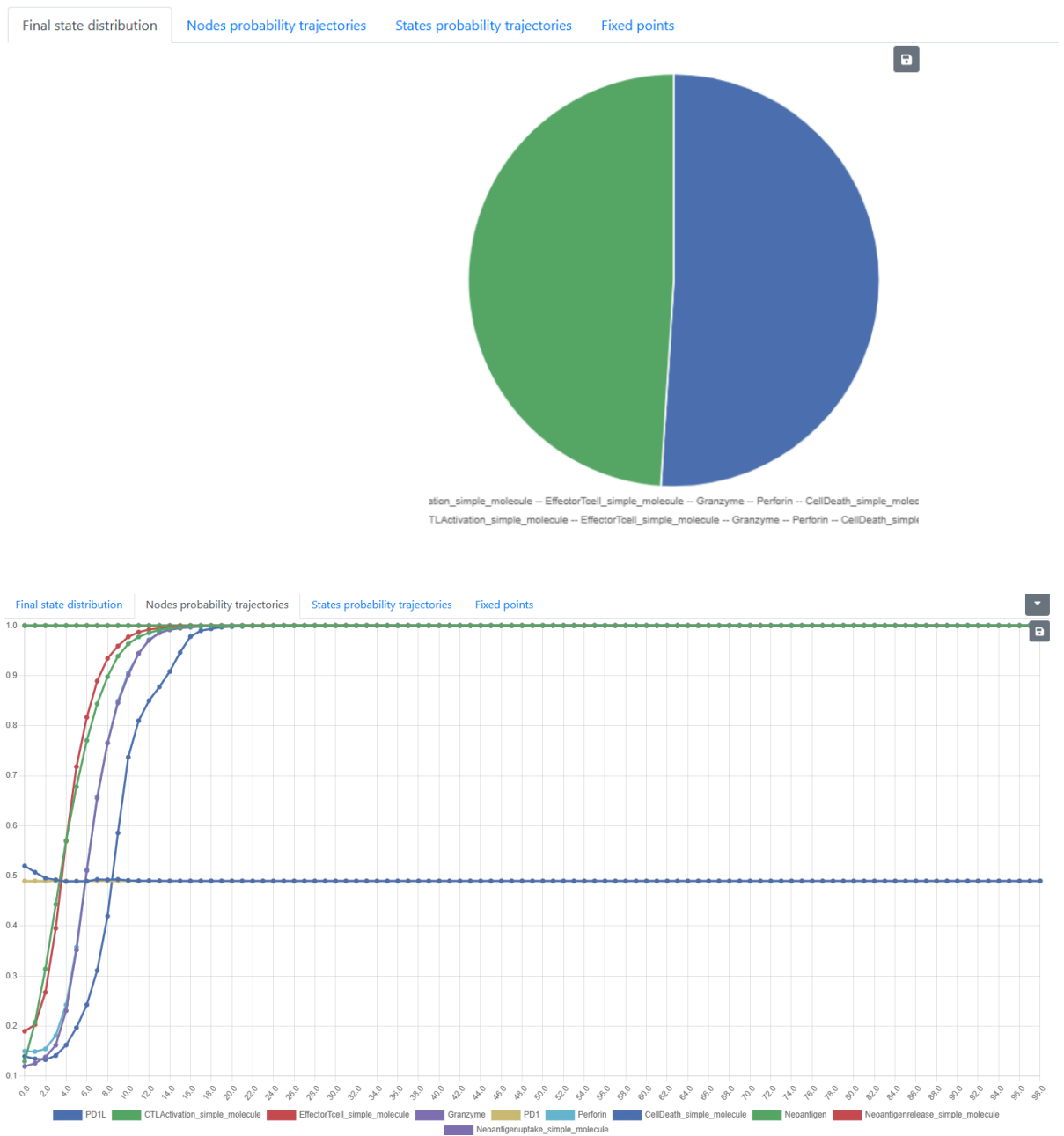
☒ virtual walls (nudge cells away from domain boundaries)





RESULTS

1. WebMaBoSS simulation:

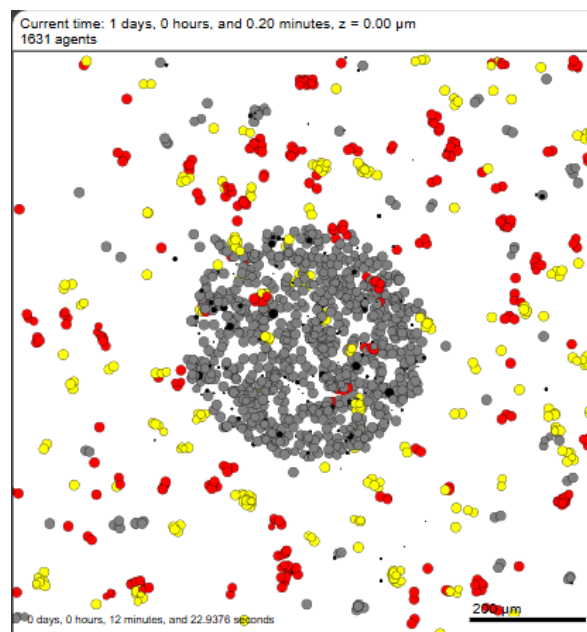
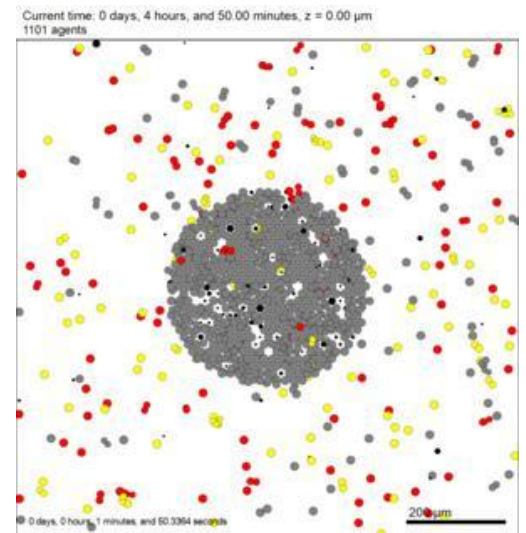
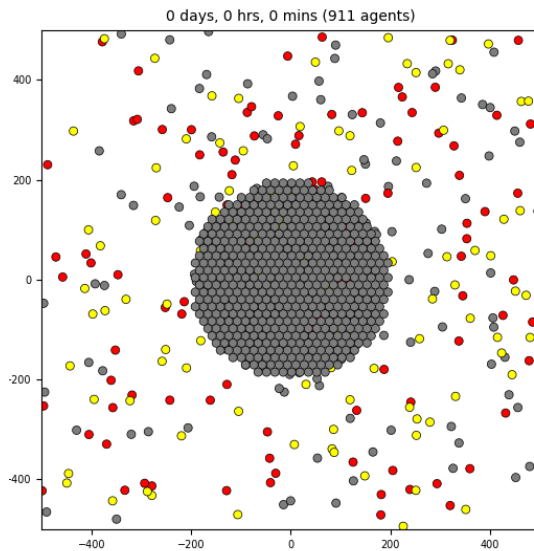


The two graphs above represent the final state distribution and the state probability trajectories.

The graph of final state distribution shows two possible final states for the system, each with about 50% probability. This suggests that the biological model we simulated can evolve into two different outcomes with equal likelihood, possibly representing two cellular fates, like survival versus cell death. The labels on the chart indicate that both outcomes involve similar molecular players (like T cell activation, granzyme, perforin, and cell death), but differ slightly in their configurations. This reflects a balanced decision-making process within the cells in the simulation.

The graph of nodal probability trajectories shows the probability trajectories of various nodes or states in a biological system over time. There are 10 different nodes represented by different coloured lines: PD1L, CTLActivation_simple_molecule, EffectorTcell_simple_molecule, Granzyme, PD1, Perforin, CellDeath_simple_molecule, Neoantigen, and Neoantigenrelease_simple_molecule. The x-axis represents time steps from 0 to 960, while the y-axis shows probability values from 0 to 1. Most nodes start at lower probability values and increase over time, eventually reaching a steady state. The PD1L node (dark blue) remains constant at around 0.5 probability throughout the entire time range. Other nodes like CTLActivation, EffectorTcell, and Granzyme show rapid increases early on before plateauing at high probabilities. The CellDeath node increases more gradually. This graph likely represents the dynamics of various components in an immune response or cell death pathway over time.

1. PhysiCell Simulation:



The simulation was run for about 24 hours. The total simulation time on the computer took about 12 minutes to simulate the interaction that occurs in a 24-hour time frame. An image of the progress was recorded at every 2-minute interval. The cancer cells undergo cell death on contact with CD8+ T cells. The mechanism of cell death starts from the outer layers and the T-cells make it into the inner layers as time passes on.

CONCLUSION

Cancer cells respond favourably on T-cell contact and cell death favours the pattern followed in the node-trajectories graph. With changes in kinetic parameters, the process can be sped up reasonably to destroy tumours, provided no mutations occur and the microenvironment conditions remain close to constant.

The pathway illustrates the process of T cell activation and its regulation. It starts off with a mature naive T cell that undergoes the antigen screening process through its TCR(T Cell Receptor) interacting with the MHC (Major Histocompatibility Complex) molecules which present antigens on its surface. This interaction then leads to the receptor activation and the receptor inhibition pathways. The receptor activation pathway involves the activation of transcription factors and AKT/mTOR signalling , ultimately leading to the CTL(cytotoxic T lymphocyte) activation and gene expression changes. On the other hand , the receptor inhibition pathway involves the interaction of PD-1 with PD-L1 , leading to the recruitment of SHP-2, a phosphatase that inhibits the downstream signalling and prevents excessive T cell activation. This process , known as immune checkpoint inhibition , helps maintain immune balance and prevents autoimmune reactions.

FUTURE SCOPE

Predictive Simulations:

Utilise the developed models to simulate how different therapies might affect individual patients based on their unique genetic and molecular profiles. By integrating patient-specific data, the model can predict how different treatment regimens could alter tumour progression and immune response, leading to more tailored and effective therapeutic approaches.

Expansion to Other Cancers:

Model Adaptation for Other Tumours: Modify and apply the existing modelling frameworks to study other types of cancers, such as breast, prostate, or colorectal cancer. Each cancer type has unique biological and molecular characteristics, and adapting the model will help understand these differences and develop specific therapeutic strategies.

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